Project Title	DeployGuard: A Reliable Repository Scanning and Deployment Orchestration System
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Module's Name	**Module Description with Deliverables**
1. User Authentication & Audit Logging	• Secure login, role-based access, and session handling are implemented. • Login activity and important user actions are recorded for auditing.
2. Project & Workspace Management	• Users can create projects, connect GitHub repositories, and select a deployment directory. • Project status, deployment history, and deployment details can be viewed from the workspace.
3. Rule-Based Stack Detection Engine	• Repository files are scanned to identify the application structure and runtime. • The required service port and basic deployment details are detected before deployment.
4. DevOps Template Engine & Pre-Flight Validation	• Deployment requirements are prepared automatically from the repository information. • Required files, commands, directory, and port settings are checked before deployment starts.
5. CI/CD Pipeline & Asynchronous Queuing	• GitHub Actions runs the build and deployment work in the background. • DeployGuard tracks each deployment operation from start to completion.
6. Security Scanning and Policy Enforcement	• Application dependencies and the built container are scanned for security issues. • Critical fixable issues can block an unsafe deployment before release.
7. Predictive Cost Analysis (FinOps)	• Estimated AWS infrastructure cost is generated for the deployment. • The cost information is saved and shown with the related project deployment.
8. Infrastructure Provisioning & Service Discovery	• Terraform provisions the AWS resources required by the application. • The deployed service endpoint is recorded and shown after successful provisioning.
9. Distributed State Management	• Deployment and infrastructure state is stored safely for reliable tracking. • Terraform state is maintained remotely so infrastructure updates stay consistent.

Module's Name	Module Description with Deliverables
10. Persistent Storage & Backup Recovery	• Persistent storage is available for applications that need saved data. • Recovery support is included so stored application data can be restored when required.
11. Zero-Downtime Container Orchestration & Event-Driven Rollbacks	• ECS manages the running application containers during releases. • A previously stable release can be restored when a deployment needs rollback.
12. Dual-Stack Monitoring & Observability	• Deployment status, application logs, and monitoring information are available in the platform. • Users can review service health and deployment activity from one place.
13. AI-Powered Log Analysis & Chatbot	• Deployment logs can be analyzed to identify common problems. • Simple troubleshooting guidance is provided to help users understand failures.
14. Alerting & Notification	• Important deployment updates and status changes are sent to users. • Notifications help users follow successful, failed, and important deployment events.
15. Infrastructure Lifecycle & Demolition (Destroy)	• Users can destroy deployed infrastructure after confirmation. • DeployGuard performs cleanup and records the result of the destroy operation.
16. Vendor Independence & Terraform Export	• Generated Terraform files can be exported from the platform. • Users can view and keep the infrastructure code outside DeployGuard.
17. Subscription & Billing	• Free and Pro subscription options are available in the platform. • Billing status and plan access are handled through the subscription flow.

Respected Evaluator, please write your remarks or suggestions in this section.	
Evaluator's Name:	**Suggestions / Remarks**

Supervisor Signature: _____

Signature FYP Committee Member: _____